Theorem: If P = NP, then EXP = NEXP.

Definitions:

EXP = ■ DTIME(2^{n^k}) and NEXP = ■ NTIME(2^{n^k}).

Proof:

EXP $\subseteq$ NEXP is trivial since deterministic machines are special cases of nondeterministic ones.

We prove NEXP $\subseteq$ EXP assuming P = NP.

Let L $\in$ NEXP. Then there exists a nondeterministic machine M and constant k such that M decides L in time $T = 2^{n^k}$.

Define A = $\{(x,1^T) : M$ accepts x within T steps$\}$.

A $\in$ NP because a certificate is the accepting computation tableau, which can be verified in polynomial time in $|(x,1^T)|$.

Since P = NP, A $\in$ P. So A can be decided in time polynomial in $|(x,1^T)|$, i.e., polynomial in T.

Given input x, compute T, construct $(x,1^T)$, and run the polynomial-time algorithm for A.

This takes time polynomial in $T = 2^{n^k}$, hence exponential in n.

Thus L $\in$ EXP. Therefore NEXP $\subseteq$ EXP.

Combining both inclusions, EXP = NEXP.

Conclusion: If P = NP, then EXP = NEXP.